Dynamic Pricing and Dispatching for Mobile Charging Services: An MDP-Based Two-stage RL Framework

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Abstract

Problem + method + experiment + main result

This paper studies ... We propose ... Numerical experiments demonstrate ...

1. Introduction

- Motivation
  - Research gap
  - why RL
  - proposed framework
  - contribution

- We formulate ...
- We develop ...
- We demonstrate ...

2. Related Work

- MCV dispatching with RL: [1]: MARL + evolutionary strategy for MCV dispatch
  - MCV pricing with RL if any
  - dual functional MCV with RL (probably none)

3. Problem description

We consider an integrated operator that manages fixed charging stations (FCSs) and a fleet of mobile charging vehicles (MCVs) to provide charging services to electric vehicles (EVs) over a planning horizon $\mathcal{T} := [T]$ consisting of T periods of equal duration Δt. The potential MCV service region is represented by a set $\mathcal{I}$ of nodes where EV charging demand may originate, with existing FCSs located at a known subset $\mathcal{J} \subseteq \mathcal{I}$. For each $(i, j) \in \mathcal{I} \times \mathcal{I}$, let $\tau_{i,j}$ denote the fixed travel time from node i to node j, which may be approximated using a distance-based metric, for example the Manhattan distance. We consider a fleet $\mathcal{M}$ of homogeneous MCVs, each with battery capacity $B > 0$. Each MCV can travel to any potential service node to provide charging services to customers' EVs or, if an FCS is present at the node, replenish its battery or discharge stored energy to the grid. Alternatively, an MCV may remain idle at its current location.

Each period consists of two sequential decision stages connected by customers' charging choices. At the beginning of each period, the operator observes the locations and battery levels of its MCVs, FCS charging and discharging prices, and potential EV charging requests. In the first stage, the operator determines the mobile charging prices at all potential service nodes. Given these prices, customers choose between receiving mobile charging services at their current locations and traveling to an accessible FCS, based on their perceived disutilities, which account for factors such as charging fees, anticipated waiting times, and travel costs. Customers who select neither option are considered lost demand. After observing the resulting demand allocation, the system proceeds to the second stage, where individual MCVs act as cooperative agents to determine their dispatching and operational decisions. Specifically, each MCV may travel to a service node to fulfill mobile charging demand, visit an FCS to recharge or discharge its battery, or remain idle at its current location, subject to operational constraints. The resulting charging services and energy transactions determine the operator's profit in the current period, while the dispatching decisions influence the system state in subsequent periods.

Pricing and dispatching decisions are closely interrelated. MCV pricing influences customers' charging choices, potentially attracting additional demand, alleviating congestion at FCSs, and increasing the operator's overall market share. The resulting demand allocation, in turn, affects MCV dispatching decisions, which determine how much of the actual mobile charging demand can be fulfilled and affect the locations and energy available to MCVs in subsequent periods. To account for these interactions, we formulate a two-stage sequential decision-making framework that integrates centralized pricing with cooperative multi-agent dispatching, with the objective of maximizing the operator's expected discounted profit over the planning horizon.

3.1 Pre-choice state and charging requests

At the beginning of each period $t \in \mathcal{T}$, the operator observes the pre-choice state $\boldsymbol{s}_t$, which includes the current period, MCV locations and battery levels, poten-

tial EV charging requests, and FCS charging and discharging prices and congestion estimates. For each MCV $m \in \mathcal{M}$, let $l_{m,t} \in \mathcal{I}$ and $e_{m,t} \in [0, B]$ denote its current location and remaining battery energy, respectively. We collect these quantities in vectors $\boldsymbol{l}_t := (l_{m,t})_{m\in\mathcal{M}}$ and $\boldsymbol{e}_t := (e_{m,t})_{m\in\mathcal{M}}$.

For each FCS $j \in \mathcal{J}$, let $p^{\mathrm{F,C}}_{j,t}$ and $p^{\mathrm{F,D}}_{j,t}$ denote its unit charging and discharging prices, respectively. We define $\boldsymbol{p}^{\mathrm{F}}_{j,t} := (p^{\mathrm{F,C}}_{j,t}, p^{\mathrm{F,D}}_{j,t})$ and collect the station prices as $\boldsymbol{p}^{\mathrm{F}}_t := \{\boldsymbol{p}^{\mathrm{F}}_{j,t}\}_{j\in\mathcal{J}}$. These prices are treated as given inputs that may vary across stations and periods. The congestion estimates at each FCS are based on potential EV charging requests and are used to determine the anticipated waiting times for customers. We assume the MCVs have access to dedicated charging and discharging facilities at FCSs, so their operations do not affect customers' waiting time.

Let $\mathcal{R}_{i,t}$ denote the set of EV charging requests arriving at demand node $i \in \mathcal{I}$ in period t. Each request $r \in \mathcal{R}_{i,t}$ specifies an energy requirement $q_r \in [\underline{q}, \overline{q}]$, where $0 < \underline{q} < \overline{q}$. For each request r, let $\mathcal{J}_{r,t} \subseteq \mathcal{J}$ denote the set of accessible FCSs. The total requested energy at node i is $\bar{d}_{i,t} := \sum_{r\in\mathcal{R}_{i,t}} q_r$. Since customers have not yet made their charging choices, $\bar{d}_{i,t}$ represents potential demand originating at node i, which may subsequently be allocated to MCV services, FCSs, or neither option.

Before posting MCV charging prices, the anticipated waiting time for mobile charging at each potential service node $i \in \mathcal{I}$ is estimated based on the current locations of MCVs. Let $\mathcal{M}_{i,t} \subseteq \mathcal{M}$ denote the set of MCVs capable of serving node i, considering their current locations, battery levels, and operational constraints. For $\mathcal{M}_{i,t} \neq \varnothing$, the anticipated waiting time is approximated by the average travel time of these MCVs to node i, given by

$$\widehat{w}_{i,t} := \frac{1}{|\mathcal{M}_{i,t}|} \sum_{m\in\mathcal{M}_{i,t}} \tau_{l_{m,t},i}. \tag{1}$$

This quantity provides a pre-dispatch estimate of the delay before actual dispatching. Mobile charging is considered unavailable at nodes with no eligible MCVs. The actual assignment of MCVs and the amount of charging demand fulfilled are determined in the second stage, after customers' charging choices are observed.

3.2 MCV pricing and customer choice

After observing the pre-choice state $\boldsymbol{s}_t$, the operator determines mobile charging prices at demand nodes where the service is available. Let $\mathcal{L}$ denote the finite set of pricing levels, where p_l is the price per kilowatt-hour (kWh) associated with level $l \in \mathcal{L}$. At each node $i \in \mathcal{I}$ with $\mathcal{M}_{i,t} \neq \varnothing$, all customers originating at the same node face an identical unit charging price $p^{\mathrm{M}}_{i,t} \in \{p_l : l \in \mathcal{L}\}$. We collect these prices in $\boldsymbol{p}^{\mathrm{M}}_t$. These prices are determined before customers making their chagring choices, whereas the FCS prices $\boldsymbol{p}^{\mathrm{F}}_t$ are given.

Each customer chooses among the charging alternatives available to their request. For request $r \in \mathcal{R}_{i,t}$, let $\mathcal{A}_{r,t}$ denote the set of available charging options, comprising the accessible FCSs $\mathcal{J}_{r,t}$, and the mobile charging option M if $\mathcal{M}_{i,t} \neq \varnothing$. The perceived costs of mobile charging at customer's current location and charging at an accessible FCS $j \in \mathcal{J}_{r,t}$ are, respectively,

$$c^{\mathrm{M}}_{r,t} := p^{\mathrm{M}}_{i,t} q_r + h^{\mathrm{M}}_{i,t} \quad \text{and} \quad c^{j}_{r,t} := p^{\mathrm{F,C}}_{j,t} q_r + h^{\mathrm{F}}_{r,j,t}. \tag{2}$$

Here, $h^{\mathrm{M}}_{i,t}$ denotes the inconvenience cost associated with the estimated MCV arrival delay $\widehat{w}_{i,t}$, while $h^{\mathrm{F}}_{r,j,t}$ represents the travel and queueing costs incurred by the customer when charging at FCS j. Both costs are expressed in monetary units and estimated using the information available before dispatching decisions are made.

Given the perceived disutilities, customers select their charging options according to a multinomial logit choice model. To account for customers who choose neither MCV nor FCS charging services, we introduce an outside option O, representing demand lost. We normalize the systematic utility of this option to zero, which serves as a reference level rather than an explicit charging cost. Specifically, the probability that customer r selects alternative $k \in \mathcal{A}_{r,t}$ is given by

$$\rho^{k}_{r,t} := \frac{\exp(-\eta c^{k}_{r,t})}{1 + \sum_{k'\in\mathcal{A}_{r,t}} \exp(-\eta c^{k'}_{r,t})}, \tag{3}$$

where $\eta > 0$ is a common cost-sensitivity parameter. Let $z^{k}_{r,t} \in \{0, 1\}$ indicate whether request r selects alternative k. Each request selects exactly one available alternative, so $\sum_{k\in\mathcal{A}_{r,t}} z^{k}_{r,t} = 1$.

Following customers' charging choices, the actual charging requests are determined. Let $\mathcal{R}^{M}_{i,t} \subseteq \mathcal{R}_{i,t}$ denote the set of requests selecting mobile charging at $i \in \mathcal{I}$, and let $\mathcal{R}^{F}_{j,t} \subseteq \mathcal{R}_{i,t}$ denote the set of requests selecting FCS $j \in \mathcal{J}$. Customers who select neither services are considered lost demand. The actual charging requests, together with their origin locations and energy requirements, provide the demand information for the subsequent MCV dispatching stage with the post-choice state $\tilde{\boldsymbol{s}}_t$.

4. Two-stage sequential RL framework

To jointly optimize dynamic MCV pricing and cooperative dispatching, we propose a Stage-Aware PPO–MAPPO (SA-PPO–MAPPO) framework that explicitly exploits the sequential structure of the two-stage decision process. In the first stage, a centralized pricing agent employs Proximal Policy Optimization (PPO) with a factorized policy to determine mobile charging prices across potential service nodes, thereby avoiding the exponential growth of the joint pricing action representation. Given these prices, customers select their charging options according to a discrete choice model, resulting in the actual charging requests. In the second stage, individual MCVs are modeled as cooperative agents and employ Multi-Agent PPO (MAPPO) with parameter sharing and centralized training with decentralized execution (CTDE) to determine

their dispatching and operational decisions. To capture the interdependence between pricing and dispatching, we introduce a stage-aware actor–critic architecture with two interconnected value functions corresponding to the pre-choice and post-choice states. These value functions enable the pricing policy to account for subsequent dispatching outcomes and the dispatching policies to account for future pricing opportunities. The pricing and dispatching policies are trained jointly using a common profit-based reward, with the objective of maximizing the operator's expected discounted profit over the planning horizon.

4.1 Stage-aware actor–critic framework

We formulate the integrated pricing and dispatching problem as a finite-horizon sequential decision process consisting of two decision stages within each period. At the beginning of period t, the operator observes the pre-choice state

$$s_t = \left(\{l_{mt}, e_{mt}\}_{m\in\mathcal{M}}, \boldsymbol{\lambda}_t, \boldsymbol{p}_t^F, \boldsymbol{w}_t^F\right),$$

where l_{mt} and e_{mt} denote the location and remaining battery energy of MCV m, respectively; $\boldsymbol{\lambda}_t$ represents potential EV charging requests; $\boldsymbol{p}_t^F$ contains the FCS charging and discharging prices; and $\boldsymbol{w}_t^F$ denotes the estimated waiting times at FCSs.

In the first stage, a centralized pricing agent determines the MCV charging prices across eligible service nodes according to

$$\boldsymbol{p}_t^M \sim \pi_{\theta_1}^P(\cdot \mid s_t), \tag{4}$$

where $\pi_{\theta_1}^P$ denotes the pricing policy parameterized by θ_1.

Given the posted MCV prices, EV customers choose among the available MCV charging services, FCS charging services, and an outside option according to the discrete choice model. The resulting charging requests are represented by

$$\boldsymbol{D}_t = \mathcal{D}\left(\boldsymbol{p}_t^M, \boldsymbol{p}_t^F, \boldsymbol{\lambda}_t, \boldsymbol{w}_t^F; \boldsymbol{\xi}_t\right), \tag{5}$$

where $\boldsymbol{\xi}_t$ captures the randomness in customers' charging choices. The actual requests consist of customers selecting MCV charging services and those selecting FCS charging services, while customers choosing the outside option do not generate charging demand for the operator. Following customer choices, the system enters the post-choice state

$$\widetilde{s}_t = \left(s_t, \boldsymbol{p}_t^M, \boldsymbol{D}_t\right). \tag{6}$$

In the second stage, each MCV $m \in \mathcal{M}$ is modeled as a cooperative RL agent. Its observation is defined as

$$o_{mt} = \left(l_{mt}, e_{mt}, \boldsymbol{p}_t^F, \boldsymbol{p}_t^M, \boldsymbol{D}_t, \boldsymbol{\zeta}_{mt}\right), \tag{7}$$

where $\boldsymbol{\zeta}_{mt}$ represents information about the other MCVs available to agent m, such as their locations and remaining battery levels. This information supports coordination among MCVs during decentralized decision-making.

Each MCV selects its operational action according to

$$a_{mt}^D \sim \pi_{\theta_2}^D(\cdot \mid o_{mt}), \qquad m \in \mathcal{M}, \tag{8}$$

where $\pi_{\theta_2}^D$ denotes the shared dispatching policy parameterized by θ_2. Since the MCV fleet is homogeneous, all agents share the same policy parameters, while their individual decisions depend on their respective observations. The dispatching actions include traveling to service nodes to provide charging services, repositioning, charging or discharging at FCSs, and remaining idle, subject to operational and battery constraints.

The two stages are coupled through the following sequential decision process:

$$s_t \xrightarrow[\text{Stage 1}]{\pi_{\theta_1}^P} \boldsymbol{p}_t^M \xrightarrow{\text{Customer choice}} \widetilde{s}_t \xrightarrow[\text{Stage 2}]{\pi_{\theta_2}^D} \boldsymbol{a}_t^D \longrightarrow (r_t, s_{t+1}), \tag{9}$$

where $\boldsymbol{a}_t^D = (a_{mt}^D)_{m\in\mathcal{M}}$ denotes the joint dispatching action, and r_t is the operator's total profit in period t. The profit includes revenues from FCS and MCV charging services and MCV energy discharge, minus energy procurement, travel, operating, and unmet-demand costs.

The objective is to jointly optimize the pricing and dispatching policies to maximize the operator's expected discounted profit:

$$\max_{\theta_1,\theta_2} J(\theta_1, \theta_2) = \mathbb{E}\left[\sum_{t=1}^{T} \gamma^{t-1} r_t\right], \tag{10}$$

where $\gamma \in (0, 1]$ denotes the discount factor.

To account for the sequential dependence between the two decision stages, we introduce two stage-specific value functions. The pre-choice value function $V^P(s_t)$ represents the expected discounted profit starting from the pricing stage, while the post-choice value function $V^D(\widetilde{s}_t)$ represents the expected discounted profit starting from the dispatching stage under the current policies. They satisfy the Bellman relations

$$V^P(s_t) = \mathbb{E}\left[V^D(\widetilde{s}_t) \mid s_t\right], \tag{11}$$

$$V^D(\widetilde{s}_t) = \mathbb{E}\left[r_t + \gamma V^P(s_{t+1}) \mid \widetilde{s}_t\right]. \tag{12}$$

5. Numerical Experiments

We conduct scenario-based numerical experiments to evaluate the proposed SA-PPO-MAPPO framework. For each scenario, the sparial and temporal distribution of potential EV requests is prescribed by $\{\boldsymbol{\lambda}_t\}_{t=1}^T$, where λ_{it} denotes the expected number of potential EV charging requests arriving at node i during period t. The realized number of potential requests is independently sampled according to $N_{it} \sim \text{Poisson}(\lambda_{it})$. The FCS charging and discharging prices fluctuate around predetermined baseline profiles according to $\{\boldsymbol{p}_t^F\}_{t=1}^T$, $p_{jt}^F = \bar{p}_{jt}^F(1 + \varepsilon_{jt})$, where $\varepsilon_{jt} \sim \mathcal{N}(0, \sigma_{jt}^2)$ or $U(-\delta_{jt}, \delta_{jt})$. Within each scenario, the demand intensity and electricity price profiles remain fixed across simulation episodes, while actual EV arrivals and electricty prices vary stochastically.

5.1 Environment

5.2 Baselines

for example fixed price + RL dispatch; dynamic pricing + heuristic dispatch; joint RL; our joint two-stage RL

5.3 Evaluation metrics

- average discounted network profit - MCV charging revenue, FCS charging revenue; V2G revenue; MCV utilization; FCS utilization; outside-option share (unserved demand), etc

5.4 Results

- compare with baselines, improves? in terms of above evaluation metrics

Table 1: Comparison of different methods.

Method	Reward	Runtime
Baseline 1	100.2	20.1
Baseline 2	112.4	24.5
Proposed	**128.6**	22.3

- Does dynamic MCV pricing create "new" demand, or mainly cannibalize the operator's own FCS demand?

- when does V2G create incremental profit for the operator, and when does it reduce the MCV fleet's ability to serve EV customers?

6. Conclusion

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References

[1] H. Li and B. Jeong, "Multiagent deep reinforcement learning with evolutionary strategy for mobile charging vehicles dispatching," IEEE Transactions on Neural Networks and Learning Systems, 2026.